Junjun Yin, Ph.D

Contact Information

Assistant Professor
Department of Data Science
Columbian College of Arts & Sciences
George Washington University

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Academic Appointments

2025- Assistant Professor, Department of Data Science, George Washington University, USA

2019-2025 Assistant Research Professor, Social Science Research Institute, Pennsylvania State University, USA

2019-2025 ICDS Associate, Institute for Computational and Data Sciences, Pennsylvania State University, USA

2016-2018 Social Science Big Data Research Scientist, Pennsylvania State University, USA

2016 Lecturer, Department of Geography and Geographic Information Science; University of Illinois at Urbana-Champaign, USA

2014-2016 Postdoctoral Research Fellow, CyberGIS Center for Advanced Digital and Spatial Studies; National Center for Supercomputing Applications; University of Illinois at Urbana-Champaign, USA

2013-2014 Assistant Lecturer, Technological University Dublin, Ireland

2007-2008 Research Assistant, Hong Kong Polytechnic University, Hong Kong SAR, China

Education

2009 - 2013 *Ph.D* in Spatial Information Science, Technological University Dublin, Ireland
Science Foundation Ireland Scholarship
Advisors: Dr. James D. Carswell (Technological University Dublin), Dr. Michela Bertolotto (University College Dublin)

2006 - 2009 *MSc* in Geoinformatics, University of Gävle, Sweden
Advisor: Dr. Bin Jiang (University of Gävle)

2002 - 2006 *BSc* in Electronics Engineering, University of Electronic Science and Technology of China

Research Interests

GIScience; Geospatial Data Science; Geographic Big Data; Human-Urban Interactions; Human Mobility; Urban Informatics; (Geo)Visual-Analytics; Spatial Data Mining and Deep Learning (GeoAI); Spatial/Graph Neural Networks; High-performance (cyberGIS), Web, and Mobile GIS

Publications

Online: Google Scholar

Submitted and In Preparation

2026 **Yin, J.**, Yin, L., Du, J., Yang, D., Hu, B. and Jiang, B., Empirical Validation of Architectural Livingness: Testing Alexander's Theory through AI-Based Measurement, Perceptual Response, and Brain Activity. (*submitted to Journal of Environmental Psychology, in review*)

2026 Chi, G., **Yin, J.**, Van Hook, J., Plutzer, E. and Xu, H., The Mis-representativeness of Twitter Data. (*submitted to Demography, in review*)

2026 Son, J.H. and **Yin, J.**, Beyond Residence: Mobility-Driven PM2.5 Exposure Disparity in the Washington, D.C. Metropolitan Area. (*submitted to Journal of Transport & Health, in review*)

2026 Lakshmanan, G.S. and **Yin, J.**, AirGuard: A Machine Learning Framework for High-Resolution PM2.5 Prediction in California Using Spatiotemporal Data Integration. (*submitted to Environmental Data Science, in review*)

2026 Hariharan, A., Pachipala, A., Saravanan, S. and **Yin, J.**, Decomposing PM2.5: The Role of Transportation in Pakistan's Air Pollution. (*in preparation, to be submitted to Atmospheric Environment*)

2026 Saravanan, S., Hariharan, A., Pachipala, A., Leiding, M., Puthal, R., Weiner, J. and **Yin, J.**, Air Pollution as a Proxy for Sectoral Economic Activity: A Scope-Conditions Framework Tested Against Algeria and Myanmar. (*in preparation, to be submitted to Economics Letters*)

2026 Charumbira (Gundani), T., Gray, J. and **Yin, J.**, Diverging trajectories of academic and policy discourse on AI governance, 2015–2025. (*in preparation, to be submitted to Patterns*)

Peer Reviewed Journals

2026 Morris, L., Chi, G., **Yin, J.**, Bodovski, Y. and Sundar, S.S., County Partisanship Affects Engagement with X (Twitter) Over the Course of a Global Pandemic. *Social Media + Society, accepted*

2025 Djenontin, I.N., Fischer, H.W., **Yin, J.** and Chi, G., 2025. Unveiling global narratives of restoration policy: Big data insights into competing framings and implications. *Geoforum*, 161, p.104241. (DOI: 10.1016/j.geoforum.2025.104241)

2025 **Yin, J.**, Brooks, M., Wang, D. and Chi, G., Characterizing Climate-Change Sentiment in Alaska with Twitter Data. *Digital Geography and Society*, 8: 100110. (DOI: 10.1016/j.diggeo.2024.100110)

2025 **Yin, J.** Unlocking the Secrets of Scenic Beauty: An artificial intelligence approach to understanding public perceptions of environmental aesthetics. *Journal of the Royal Society Interface*, 22(225): 20250045. (DOI: 10.1098/rsif.2025.0045)

2025 Sundar, S.S., Snyder, E., Liao, M., **Yin, J.**, Wang, J., and Chi, G., Sharing without clicking: The concerning phenomenon of mindless news forwarding on social media. *Nature Human Behaviour*, 9(1), pp. 156-168. (DOI:10.1038/s41562-024-02067-4)

2024 Chi, G., Zhou, S., Mucioki, M., Miller, J., Korkut, E., Howe, L., **Yin, J.** et al. "Climate impacts on migration in the Arctic North America: existing evidence and research recommenda-

tions." *Regional Environmental Change*, 24(2): 47

2023 **Yin, J.** and Chi, G. (2023), A Tale of Three Cities: Uncovering human-urban interactions with geographic-context aware social media data. *Urban Informatics, 1(1), 20* (DOI:10.1007/s44212-022-00020-2)

2023 Yu, M., Zhang, S., Zhang, K., **Yin, J.**, Varela, M. and Miao, J. (2023), Developing high-resolution PM2.5 exposure models by integrating low-cost sensors, automated machine learning, and big human mobility data. *Frontiers in Environmental Science* (DOI:10.3389/fenvs.2023.1223160)

2023 Liang, Y., **Yin, J.**, Park, S., Pan, B., Chi, G., and Miller, Z. (2023), Using social media user profiles to identify visitor demographics and origins in Yellowstone national park. *Journal of Outdoor Recreation and Tourism* (DOI:10.1016/j.jort.2023.100620)

2022 **Yin, J.**, Gao, Y. and Chi, G. (2022), An evaluation of geo-located Twitter data for measuring human migration. *International Journal of Geographical Information Science*, 36(9), pp. 1830-1852, DOI: 10.1080/13658816.2022.2075878

2022 Li, C., Zhao, J., **Yin, J.** and Chi, G. (2022), Park access affects physical activity: new evidence from geolocated Twitter data analysis. *Journal of Urban Design* (DOI: 10.1080/13574809.2022.2118698)

2022 Alba, C., Pan, B., **Yin, J.**, Rice, W., Lin, M., Liang, Y. and Mitra, P. (2022), COVID-19's impact on visitation behavior to US national parks from communities of color - evidence from mobile phone data. *Scientific Reports*, 12:13398, DOI: 10.1038/s41598-022-16330-z

2022 Liang, Y., **Yin, J.**, Pan, B., Lin, M., Miller, L., Taff, D., Chi, G. (2022), Assessing the Validity of Mobile Device Data for Estimating Visitor Demographics and Visitation Patterns in Yellowstone National Park. *Journal of Environmental Management*, 317(115410), DOI:10.1016/j.jenvman.2022.115410

2022 Chi, G., **Yin, J.**, Morris, L., and Bodovski, Y., Global Tweet Mentions of COVID-19 (2022). *Health Data Science*

2021 **Yin, J.** and Chi, G., Characterizing People's Daily Activity Patterns in the Urban Environment: A mobility network approach with geographic context-aware Twitter data. *Annals of the American Association of Geographers*, 111(7), pp.1967-1987

2020 Abdar, M., Basiri, M.E., **Yin, J.**, Asadi, S. and Chi, G. (2020), Energy Choices in Alaska: Mining People's Perception and Attitudes from Geotagged Tweets. *Renewable & Sustainable Energy Reviews*, 124, p.109781

2019 Pu, Y., Zhao, X., Chi, G., Zhao, S., Wang, J., Jin, Z. and **Yin, J.** (2019). Design and implementation of a parallel geographically weighted k-nearest neighbor classifier. *Computers & Geosciences*, 127, pp. 111-122

2018 Gao, Y., Wang, S. Padmanabhan, A., **Yin, J.** and Cao, G. (2018). Mapping Spatiotemporal Patterns of Events Using Social Media: A Case Study of Influenza Trends. *International Journal of Geographical Information Science*, 32(3), pp. 425-449

2017 **Yin, J.**, Soliman, A., Yin, D. and Wang, S. (2017). Delineate Urban Boundaries in Great Britain from the Network of Large Scale Twitter User Spatial Interactions. *International Journal of Geographical Information Science*, 31(7), pp. 1293-1313

2017 Soliman, A., Soltani, Q., **Yin, J.**, Padmanabhan, A., and Wang, S (2017). Social sensing of urban land use based on analysis of Twitter users' mobility patterns. *PLoS ONE*, 12(7): e0181657. DOI:10.1371/journal.pone.0181657

2017 Zheng, K., Kwan, M.P., Fang, L., **Yin, J.**, Gu, D. and Fu, Y. (2017). A Topology-concerned

Spatial Vector Data Model for Column-oriented Databases. *International Journal of Database Theory and Application*, 10(5), pp. 33-46

2016 **Yin, J.**, Gao, Y., Du, Z. and Wang, S. (2016). Exploring Multi-Scale Spatiotemporal Twitter User Mobility Patterns with a Visual-Analytics Approach. *ISPRS International Journal of Geo-Information*, 5(10):187.

2016 Jiang, B., Ma, D., **Yin, J.** and Sandberg, M. (2016). Spatial Distribution of Tweet Numbers and Densities in Cities. *Geographical Analysis*, 48(3), pp. 337-351

2015 Jiang, B.,**Yin, J.** and Liu, Q. (2015). Zipf's Law for All the Natural Cities around the World. *International Journal of Geographical Information Science, 29(3), pp. 498-522*

2014 Jiang, B. and **Yin, J**. (2014). Ht-Index to Quantify the Fractal or Scaling Structure of Geographic Features. *Annals of the Association of American Geographers*, 104(3), pp. 530–540

2013 **Yin, J.** and Carswell, J.D. (2013). Spatial Search Techniques for Mobile 3D Queries in Sensor Web Environments. *ISPRS International Journal of Geo-Information*, 2(1): pp.135-154

2010 Carswell, J. D., **Yin, J.** and Gardiner, K. (2010). 3DQ: Threat Dome Visibility Querying on Mobile Devices *GIM International*, 24(8)

2010 Carswell, J.D., Gardiner, K. and **Yin, J.** (2010). Mobile Visibility Querying for LBS. *Transactions in GIS*, 14(6): pp. 791-809, Wiley online library

2009 Jiang, B., **Yin, J.** and Zhao, S. (2009). Characterizing the human mobility pattern in a large street network. *Physical Review E*, 80(2), 021136

2008 Jiang, B., Zhao, S. and **Yin, J.** (2008). Self-organized natural roads for predicting traffic flow: a sensitivity study. *Journal of Statistical Mechanics: Theory and Experiment*, P07008, IOP Publishing

Peer Reviewed Conferences and Lecture Notes

2023 **Yin, J.** and Chi, G. (2023). Fine-grained Population Estimates: Infusing Deep Learning into Dasymetric Mapping of Buildings. *2023 North American Meetings of the Regional Science Association International*, San Diego, CA, November 15 -18, 2023

2022 Miller, J., **Yin, J.** and Chi, G. (2022). Does Environmental Perception Matter? Using Twitter Data to Explore the Mediating Effects of Environmental Risk Perception on Out-Migration From Coastal Alaska. *Population Association of America (PAA) Annual Meeting*, April 6-9, 2022 (full paper, oral presentation)

2022 Pan, B., Savanapelli, V., Shukla, A., **Yin, J.**∗ (2022). Monitoring Human-Wildlife Interactions in National Parks with Crowdsourced Data and Deep Learning. *Proceedings of the ENTER 2022 eTourism Conference*, January 11-14, 2022 (∗*corresponding author, ∗Best paper award*)

2021 Liang, Y., **Yin, J.**, Lin, M., Pan, B., Chi, G. (2021). Assessing the validity of SafeGraph data for visitor monitoring in Yellowstone National Park. *Travel and Tourism Research Association: Advancing Tourism Research Globally*, June 14-16, 2021, Fort Worth, Texas, USA

2021 **Yin, J.**, Brooks, M., Wang, D. and Chi, G. (2021). Characterizing Climate-Change Sentiment in Alaska with Twitter Data. *Population Association of America (PAA) Annual Meeting*, May 5-8, 2021 (full paper, oral presentation)

2021 **Yin, J.**, Gao, Y., Chi, G. and Van Hook, J. (2021). An Evaluation of Geo-located Twitter Data for Measuring Human Migration. *Population Association of America (PAA) Annual Meeting*, May 5-8, 2021 (full paper, oral presentation)

2020 Liang, Y., **Yin, J.**, Pan, B., Chi, G., Andris, C., Miller, Z., Jorgenson, J. and Nickerson, N. (2020). Understanding Demographics and Experience of Tourists in Yellowstone National Park through Social Media. Travel and Tourism Research Association: Advancing Tourism Research Globally. 54

2019 **Yin, J.**, and Chi, G. (2019). Understanding Spatiotemporal Urban Activity Patterns with Geo-located Twitter Data: A GIS-based Synthesis Approach. *66th Annual North American Meetings of the Regional Science Association International (NARSC)*, November 13-16, 2019, Pittsburgh, Pennsylvania

2019 **Yin, J.**, and Chi, G. (2019). An Evaluation of Geo-located Twitter Data as Indicators for Human Migration. *IUSSP Research Workshop on Digital Demography in the Era of Big Data*, June 6-7, 2019, Seville, Spain

2019 Chi, G., **Yin, J.**, Van Hook, J., Plutzer, E. and Heng, X. (2019). The Generalizability of Twitter Data for Population Research. *Population Association of America (PAA) Annual Meeting*, April 22-25, Washington, DC, USA

2018 **Yin, J.**, Chi, G. and Van Hook, J. (2018). Evaluating the Representativeness in the Geographic Distribution of Twitter User Population. *The 12th Workshop on Geographic Information Retrieval in ACM SIGSPATIAL 2018*, November 6-9, 2018, Seattle, Washington, USA

2018 Jeong, M., **Yin, J.**, and Wang, S. (2018). Outliers Detection and Comparison of Origin-Destination Flows with Data Depth. *The 10th International Conference on Geographic Information Science (GIScience 2018)*, August 28-31, 2018, Melbourne, Australia

2017 **Yin, J.** (2017). Mining sequential mobility pattern from semantics enriched Twitter user trajectories. *NSF Mobility Workshop on Analyzing Movement and Mobility within Geographic Context*, May 11-12, 2017, The Ohio State University, Columbus, Ohio

2017 Chi, G., **Yin, J.** and Hook, J.V. (2017). Predicting Twitter User Demographics as a First Step in Big Data for Population Research. *The 28th International Population Conference of the International Union for the Scientific Study of Population*, October 29-November 4, 2017, Cape Town, South Africa

2016 **Yin, J.**, Lu, B., Yin, D. and Wang, S. (2016). A scalable visual-analytics approach for studying mobility networks: Revealing hierarchical structures in taxi mobility flows of New York, *The Third International Conference on CyberGIS and Geospatial Data Science*, July 26-28, 2016, Urbana, Illinois

2016 **Yin, J.**, Gao, Y. and Wang, S. (2016). Urban Sensing from Volunteered Citizen Participation using Mobile Devices. In *Seeing Cities through Big Data: Research, Methods and Applications in Urban Informatics*, Springer

2015 Soliman, A., **Yin, J**, Soltani, K., Padmanabhan, A., and Wang, S. (2015). Where Chicagoans tweet the most: Semantic analysis of preferential return locations of Twitter users, *1st International Workshop on Smart Cities and Urban Analytics 2015, 23rd ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*

2014 **Yin, J.** and Wang, S. (2014). Understanding the Evolvements of Natural Cities from Nighttime Light Images: A CyberGIS-Enhanced Approach to Large Scale Geospatial Data Analysis, *The Second International Conference on CyberGIS and Geodesign*, August 19-21, 2014, Redlands, California

2014 **Yin, J.**, Gao, Y. and Wang, S. (2014). CyberGIS Enabled Urban Sensing from Volunteered Citizen Participation using Mobile Devices. *NSF Workshop on Big Data and Urban Informatics 2014*

2013 Truong-Hong, L., Thi, T.T.P, **Yin, J.** and Carswell, J.D. (2013). Detailed 3D building models for Google Earth integration. In *Proceedings of the 13th International Conference on Computational Science and Its Applications (ICCSA 2013)*, Ho Chi Minh City, Vietnam: Springer (∗***Best paper award***)

2013 Thi, T.T.P, Truong-Hong, L., **Yin, J.** and Carswell, J.D. (2013). Exploring Spatial Business Data: A ROA-based eCampus application. In *Proceedings of the 11th International Conference on Web and Wireless Geographical Information Systems*, Banff, AB: Springer Berlin Heidelberg, pp. 164-179

2012 **Yin, J.** and Carswell, J.D. (2012). Effects of Variations in 3D Spatial Search Techniques on Mobile Query Speed vs Accuracy. *Web and Wireless Geographical Information Systems*, Naples, Italy. (∗***Best paper award***)

2012 Carswell, J.D. and **Yin, J.** (2012). Mobile Spatial Interaction in the Future Internet of Things. In *Proceedings of the 20th International Conference on Geoinformatics (GEOINFORMATICS)*, 2012, pp. 1-6

2012 **Yin, J.** and Carswell, J.D. (2012). MobiSpatial: Open-source for Mobile Spatial Interaction. *Proceedings of the 27th Annual ACM Symposium on Applied Computing*, pp. 572-573

2011 **Yin, J.** and Carswell, J.D. (2011). Touch2Query enabled mobile devices: A case study using OpenStreetMap and iPhone. *Web and Wireless Geographical Information Systems*, pp. 203-218, Springer

2009 Gardiner, K., **Yin, J.** and Carswell, J.D. (2009). EgoViz: A mobile-based spatial interaction system. *Web and Wireless Geographical Information Systems*, pp. 135-152, Springer

Books

2009 **Yin, J.** (2009), *The Topological Patterns of Urban Street Networks: Exploring the topological patterns of urban street networks from analytical and visual perspectives*, VDM: Germany, ISBN: 3639161734

Data

2020 **Yin, J.**. Estimation of Twitter user demographics in the USA, 2014. *Harvard Dataverse*, doi:10.7910/DVN/PKKAPK

Invited Talks

2024 **Yin, J.** (2024). Modeling Geo-Complexity in Human-Urban Interactions: Computational Geography meets Geographic Big Data. June 2024, Department of Geography, the University of Manchester, United Kingdom

2024 **Yin, J.** (2024). A New Era of Urban Informatics: Multi-Layered Data Synthesis Approaches and Frameworks. June 2024, Department of Geography and Environmental Science, the University of Southampton, United Kingdom

2023 **Yin, J.** (2023). Geographic Data Science: Spatial Data mining and machine learning with mobility Big Data. April, 2023, Department of Geography, the University of Liverpool, United Kingdom

2022 **Yin, J.** (2022). A computational geography approach to understanding human-urban environment interactions. July 2022, Department of Land Surveying and Geo-Informatics (LSGI), the Hong Kong Polytechnic University, Hong Kong SAR, China

2021 **Yin, J.** (2021). Human-Urban Interactions: A computational geography approach. October 2021, Department of Geography, Pennsylvania State University, University Park, PA, USA

2020 **Yin, J.** (2020). Uncovering universal human activity patterns with geospatial big data: complexity, unity, and regularity. March 2020, Auburn University, Auburn, AL, USA

2020 **Yin, J.** (2020). Computational Social Sciences: spatial interactions and geo-complexity. February 2020, University of Arkansas, Fayetteville, AR, USA

2019 **Yin, J.** (2019). Spatial Networks: A computational geography approach to new insights into spatial interactions and geo-complexity. January 2019, University of Florida, Gainesville, FL, USA

2018 **Yin, J.** (2018). Computational Geography for Capturing Geo-Complexity in Urban Studies. June 2018, Newcastle University, Newcastle upon Tyne, UK

2018 **Yin, J.** (2018). Advanced Methods and Techniques for Big GeoData. June, 2018, ITC, University of Twente, Enschede, the Netherlands

2018 **Yin, J.** (2018). Geo-Complexity and Human mobility: Through the lens of spatial Big Data to understand urban dynamics. February, 2018, University of Denver, Denver, USA

2018 **Yin, J.** (2018). High-performance Computing with Hadoop. Software in the Humanities and Social Sciences Workshop. February, 2018, Penn State, University Park, USA

Presentations

2018 **Yin, J.** (2018). Spatial Interaction Patterns of Preferential Return Behaviors in People's Daily Life. The Association of American Geographers Annual Meeting, April 10-14, 2018, New Orleans, Louisiana, USA

2017 **Yin, J.** (2017). A mobility network approach to modeling urban spatial interactions: Insights from the movement Big Data in New York City. The Association of American Geographers Annual Meeting, April 5-9, 2017, Boston, Massachusetts, USA

2016 **Yin, J.** and Wang, S. (2016). Mining Mobility Patterns From Semantic Twitter User Trajectories. The Association of American Geographers Annual Meeting, March 29-April 2, 2016, San Francisco, California, USA

2015 **Yin, J.** and Wang, S. (2015). Finding community structures of UK cities based on large-scale Twitter user mobility patterns, *Association of American Geographers Annual Meeting, Chicago, 2015*, April 21-25, 2015, Chicago, Illinois, USA

2011 **Yin, J.** (2011). Web-service based Mobile Geospatial Application Development using Python, *PyCon 2011*, Dublin, Oct 8-9

Research Grants and Projects

PI. "Deep learning for Fine-grained Population Estimation in the Arctic Region". Nvidia Academic Hardware Grant Program. Project dates: 06/01/2023–05/30/2024

Co-PI. "RAPID: Using Mobile Phone Data to Understand the Impacts of the COVID-19 Pandemic on Food Assistance Use in Alaska". NSF (National Science Foundation) SOC

grant, $200,000. Project dates: 02/01/2022–01/31/2025

PI. "A Deep Learning-enabled Dasymetric Mapping Approach for Fine-Scale Population Estimation in Alaska". ICDS computational and Data Sciences seed grant award, $35,000 Project dates: 05/01/2020–03/30/2022

PI. "Mobility Patterns and Their Influence on Food Security in Alaskan Communities: Using Digital Trace Data for Social and Behavioral Research during the COVID-19 Pandemic". SSRI COVID-19 Pandemic Emergency Research Funding, $30,000 Project dates: 04/01/2020–03/30/2022

PI. "Identifying Discrimination During COVID-19 Pandemic on Twitter", ICDS (Institute for Computational and Data Sciences) COVID-19 explore grant, $5,840. Project dates: 07/01/2020–06/30/2022

Co-PI. "Are we more willing to speak up through mobile phones? A comparison of desktop vs. mobile political sharing on Facebook". SSRC (Social Science Research Council) grant, $49,017. Project dates: 02/01/2020–01/31/2021

Senior Personnel. "Pursuing Opportunities for Long-term Arctic Resilience for Infrastructure and Society". NSF (National Science Foundation) NNA grant, $3,000,000. Project dates: 01/01/2020–12/31/2022

Co-PI. "RR: The Generalizability and Replicability of Twitter Data for Population Research". NSF (National Science Foundation) SOC grant, $500,000. Project dates: 07/15/2018–06/30/2021

Co-PI. "Understanding National Park Visitors' Spatial Behavior with Twitter Data". Penn State Seed Grant (SSRI, ICS, and IST), $20,000. Project dates: 03/01/2019–02/28/2021

PI. "Data Science: Mining sequential mobility patterns from semantic Twitter user trajectories". Microsoft Azure Data Science Research Award, $10,000. Project dates: 05/01/2018–04/30/2019

Co-PI. "An Innovative Approach to Tackle the Opioid Epidemic: Utilizing Twitter Data and Integrating Big Data Analytics and Spatial and Social Network Analyses'. Social Science Research Institute Level 2 Research Award, $20,000. Project dates: 09/01/2018–08/30/2020

Co-I. "Cognitive Changes Associated with Hormonal Treatment for Breast Cancer". Health & Environment initiative seed grant award, $49,972. Project dates: 07/01/2017–06/30/2019

Co-PI. "The Generalizability and Replicability of Twitter Data for Population Research". ICS seed grant award, $35,000. Project dates: 07/01/2018–06/30/2019

Co-PI. "Ecological Migration in a Large-Scale Quasi-Experiment Design in China: Implications of Climate Change, Landscape Structure, Ecosystem Services and Government Intervention". IEEE seed grant award, $25,000. Project dates: 04/15/2017–06/30/2018

PI. "Mining Twitter User Demographics as a First-Step in Big Data for Population Research". XSEDE (Extreme Science and Engineering Discovery Environment) Startup computational resource allocation award, 50,000 SUs & 40 TB storage (estimated $2,794.78). Project dates: 03/27/2017–03/26/2019

PI. "A cloud computing enabled GIS platform for the integration and synthesis of multi-layer geospatial data sources in urban studies: Understanding urban dynamics from geospatial Big Data". Microsoft Azure Data Science Research Award, $20,000. Project dates: 12/04/2016–12/03/2017

SP/Research Scientist. "Activity space contexts and measuring environmental exposure

in behavioral research". R21 NCI grant (ASCMEE Study).

Honors and Awards

2013 *Fiosraigh Head of School Research Award*: for excellence in research, Technological University Dublin

2013 *EU Future Internet Award*: for excellence in Future Internet research

2009 *PhD scholarship from Science Foundation Ireland (SFI)*: StratAG PhD scholarship

2006 *Outstanding Graduate Student Award*: University of Electronic Science and Technology of China

2006 *Best Undergraduate Dissertation Award*: Department of Electronic Engineering, University of Electronic Science and Technology of China

2005 *National Undergraduate Electronic Design Contest: Second prize*:Embedded system for Multichannel and frequency wave generating, (Team: Junjun Yin, Yu Chen, and Xia Yu)

Teaching

Graduate Courses, George Washington University

2026- Visualization of Complex Data (DATS 6401), *Lecturer*

2026- Data Science Capstone (DATS 6501), *Lecturer*

Undergraduate Courses, George Washington University

2026- Data Warehousing (DATS 2104), *Lecturer*

2025- Data Visualization for Data Science (DATS 2102), *Lecturer*

Earlier Teaching/Instructing

2024 Stochastic Modeling and Computation (SMAC series): Geographic Data Science, *Guest lecturer*, Penn State

2023 SoDA501 : Social Data Analytics, Spring, *Guest lecturer*, Penn State

2022 Quantitative Developmental Systems Methodology (QuantDev Series): Machine learning/Deep learning for text analysis in social science applications, *Guest lecturer*, Penn State

2021 Spatial Networks, *lecturer*, Penn State

2017 Computational Social Sciences: data, methods, and applications, *lecturer*, Penn State

2016 GEOG 479: Advanced Topics in GIS – CyberGIS, Spring, 2016, *Lecturer*, Department of Geography and Geographic Information Science, UIUC

2015 Parallel Databases, *workshop/short course series*, CyberGIS Center, UIUC

2015 Interactive Visualization of Large-scale Movement Data using Apache Spark, *Summer school series*, CyberGIS Center, UIUC

2015 Introducing the CyberGIS Toolkit in high performance computing environment, *workshop/short course series*, CyberGIS Center, UIUC

2015 Taming with geospatial Big Data with Hadoop, *workshop/short course series*, CyberGIS Center, UIUC

2014 Getting to know CyberGIS, *workshop/short course series*, CyberGIS Center, UIUC

2014 - 2016 Education and outreach at the CyberGIS Commons, *Lead*, University of Illinois at Urbana-Champaign, 2014 - 2016

2013 Programming in C, *assistant lecturer*, Technological University Dublin, 2013 - 2014

2013 Object oriented programming for game development, *senior demonstrator/assistant lecturer*, Technological University Dublin, 2013 - 2014

2013 User interface design and GUI programming with Java, *senior demonstrator/assistant lecturer*, Technological University Dublin, 2013 - 2014

Mentorship

2026 Project: Testing whether LLM-rated architectural properties predict crowd-sourced scenicness (800 UK images, 212,000 ratings). Capstone project, continued as volunteer research assistant from October 2026 to validate the method, extend it to U.S. imagery, and prepare a manuscript. M.S. student: Aniruddh Rajagopal

2026 Project: Machine-learning prediction of high-resolution PM2.5 in California's South Coast Air Basin. Capstone project, continued as volunteer research assistant; first author of the resulting AirGuard manuscript (submitted to Environmental Data Science, in review). M.S. student: Gowri Lakshmanan (M.S. 2026)

2025- Academic advisor for Data Science undergraduate majors (last names Q–Z), George Washington University

2023 Project: Mobility Big Data for studying return migrations amid natural disasters and environmental change. Ph.D student: Yining Feng

2023 Drawdown (Climate Change) Scholar mentor. Project: Geospatial Analysis of Socioeconomic Factors Impacting Electric Vehicle Charging Station Accessibility. Student: Fazil Farhan Iqbal

2022 Project: Exploring national park visitor dynamics with Location Based Social Media and Mobile Location Data. Ph.D student: Yun Liang (Currently Teaching Assistant Professor at Kansas State University)

2021 Capstone program mentor. Project: Assessing the impact of the COVID-19 pandemic on national park visitor patterns with model-driven approaches and mobility Big Data. Undergraduate student: Charles Alba (Currently Ph.D student at the Washington University in St Louis)

Service

University Service

- **Member** GPAC Committee, general education curriculum and course designations, Columbian College of Arts & Sciences, George Washington University (September 2026–present)
- **Member** Research Advisory Council, Columbian College of Arts & Sciences, George Washington University (2026–present)

Editorial Board

- **Editorial board member** Journal of Geography and Cartography
- **Topic editor** Geosciences
- **Guest editor** Special Issue of "Spatial Demography": Big Spatial Data in Demography (2023)
- **Guest editor** Special Issue of "Chinese Sociological Dialogue": Spatiotemporal Big Data and Sustainable Social Development (2017)

Program Committees and Organizers

- **PC** member for W2GIS 2020: 18th International Symposium on Web and Wireless Geographical Information Systems, Wuhan, China, May 14-15, 2020
- **Chair** for Innovations in Spatial Data Analysis and Modeling, 66th Annual North American Meetings of the Regional Science Association International (NARSC), Pittsburgh, Pennsylvania, November 13-16, 2019
- **PC** member for W2GIS 2019: 17th International Symposium on Web and Wireless Geographical Information Systems, Kyoto, Japan, May 16-17, 2019
- **PC** member for International Conference on Location-based Social Media Data, March 5-7, 2015, Athens, Georgia, USA
- **Co-chair** for Advances in Spatial Interaction Models and Methods in the Big Data Era I, Spatiotemporal Symposium, 2018 AAG Annual Meeting, New Orleans, April 10 – April 14, 2018
- **Chair** for Big Movement Data for Geospatial Analytics on Urban Interactions, Symposium on Human Dynamics in Smart and Connected Communities, 2017 AAG Annual Meeting, Boston, Massachusetts, April 5 - April 9, 2017
- **Chair** for Understanding Urban Dynamics Based on Movement Big Data, 2016 AAG Annual Meeting, San Francisco, California, March 29 - April 2, 2016

Reviewer for Proposals

- ***Methodology, Measurement, and Statistics Program**, National Science Foundation* (2019)

Reviewer for Journals

Reviewed articles from over **50** journals, including Annals of the American Association of Geographers, International Journal of Geographical Information Science, and Applied Geography, etc.

Skills

Programming languages: Python, R, Java, Objective-C, Matlab, JavaScript, NoSQL
Neural nets: ResNet, U-net, Convolutional Neural Networks, Graph Neural Networks
High performance computing (HPC): Hadoop, Spark, MongoDB
Mobile and web geo-visualization: iOS and Android, Cesium 3D Globe, D3.js
Spatial databases: PostGIS, Oracle Spatial, SpatiaLite, ArcGIS